Miquel Obrador-Reina

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Education

Universitat Politècnica de València, PhD in Computer Science – Valencia, Spain — Sept 2024 – present

- Second-year PhD student at the PRHLT Research Center.
- Research focus on efficient Deep Learning, Vision-Language Models (VLM), and Model Compression.

Universitat Politècnica de València, MS in Computer Science – Valencia, Spain — Sept 2023 – July 2024

Universitat Politècnica de València, BS in Computer Science – Valencia, Spain — Sept 2019 – July 2023

Experience

PRHLT Research Center - Universitat Politècnica de València, Predoctoral Researcher – Valencia, Spain — Sept 2024 – present

- Researching efficient Vision-Language Models (VLLMs) through dynamic token pruning and model compression techniques.
- Investigating SVD-based compression and gradient-based refinement for Large Language Models to enable deployment on consumer-grade hardware.

Publications

Gradient-Based Refinement of SVD Factorizations via Layer-Sequential Distillation in LLMs (GLRD) — Dec 2024
Miquel Obrador-Reina
(arXiv:2412.xxxx [cs.LG])

Unveiling Differences: A Vision Encoder-Decoder Model for Difference Medical Visual Question Answering — May 2025
Luis-Jesus Marhuenda, Mohamed Aas-Alas, Miquel Obrador-Reina, Alberto Albiol, Roberto Paredes
(MIDL 2025 (Oral Presentation))

Projects

Dynamic Token Pruning for VLLMs — 2024 - present

- Engineered a dynamic image patch selection mechanism for Large Vision-Language Models using a lightweight text-encoder.
- Optimized computational efficiency in VLLMs by identifying and pruning semantically redundant image tokens during inference without sacrificing accuracy.

GLRD - Gradient-Based LLM Compression — 2025 - 2026

- Developed a post-training compression framework (GLRD) that surpasses traditional SVD methods through layer-sequential distillation.
- Proved feasibility of compressing 13B parameter LLMs on consumer-grade hardware (single NVIDIA RTX 4090 24GB).

Skills

Programming: Python, Bash, Java, MATLAB/Octave, LaTeX

AI Frameworks: PyTorch, Hugging Face (Transformers, Accelerate, Diffusers), Vision Transformers (ViT)

Tools & DevOps: Git, GitHub Actions, Docker, Slurm, uv, Conda

Systems & Hardware: Linux (Ubuntu, Kubuntu, Fedora), SSH/Remote development, VRAM Optimization

Interests: Music Production, Bass Guitar